

Which resonance does an AGN-disk companion hit first?

Implementation note for `src/GW_forward/analysis/`:
`agn_cloud_resonances.py`, `agn_julia_pipeline.py`, `evolve_agn_history.jl`,
`run_agn_scan.sh`, `agn_scan_summary.py`

1 Question

A massive black hole (MBH) grows by episodic thin-disk accretion and hosts a scalar cloud of mass μ . During an accretion episode a stellar-mass BH forms in or is captured by the disk, migrates inward on a circular, equatorial, prograde orbit, and sweeps through the gravitational-atom resonances. We record the *first* transition it crosses, as a function of the boson mass, for MBHs that end up in the LISA band. The pipeline has three stages: a cheap accretion history without the boson, the full cloud evolution along that history (Julia), and a Monte Carlo over companions.

2 Accretion histories

Seeds form at $z = 15$ (flat Λ CDM, $H_0 = 67.7$, $\Omega_m = 0.31$) with M_{seed} log-uniform in $[10^2, 10^5] M_\odot$ and $a_{\text{seed}} \in [0, 0.5]$. Accretion is ON in episodes of fixed length $t_{\text{ep}} = 10$ Myr at a duty cycle $f \sim U[0.01, 0.05]$ per MBH: $n_{\text{ep}} = \text{round}(fT/t_{\text{ep}})$ episodes, one per slot of length T/n_{ep} with a uniform start inside it (T = time from the seed to today). While ON,

$$\frac{d \ln M}{dt} = \frac{\lambda_{\text{on}}(1 - \epsilon)}{\epsilon t_{\text{Edd}}}, \quad \frac{da}{d \ln M} = \frac{L_{\text{isco}}(a)}{E_{\text{isco}}(a)} - 2a, \quad (1)$$

with $\lambda_{\text{on}} = 0.3$, $\epsilon = 0.1$, $t_{\text{Edd}} = \sigma_{TC}/4\pi G m_p \simeq 450$ Myr, coherent (prograde) Bardeen spin-up capped at $a = 0.998$. A history is kept only if $10^5 \leq M(z=0)/M_\odot \leq 10^7$, evaluated *without* superradiance. The total growth is a factor 2–60, so about 35% of the draws are kept and the kept seeds are heavy ($10^{3.6}$ – $10^5 M_\odot$); light seeds need a larger λ_{on} or duty cycle. Without the boson every kept MBH reaches $a = 0.998$.

3 Cloud evolution

`evolve_agn_history.jl` cuts each history into its ON and OFF segments and evolves the BH and the cloud through each with one `solve_system` call: $\dot{M} = (d \ln M / dt) M$ with Bardeen spin-up while ON, no accretion while OFF, and superradiance, scattering (decay constant f_a) and GW emission throughout, for levels up to N_{max} (`track_alpha` on, evolving 211 cap). The output is $M(t)$, $a(t)$ and $M_{c,nlm}(t)/M$ from the seed to today. Superradiance makes the MBH up to $\sim 30\%$ lighter today than the boson-free estimate used for the band cut (`analyze --band_check` drops histories that leave the band).

4 Companions

For each evolved history, n_{inj} companions ($m = 10 M_\odot$) are placed at times uniform in the ON time with $z < 1$. Each has a formation channel, disk capture ($r_0 \in [10^3, 10^6] r_g$) or in-situ formation ($r_0 \in [10^5, 10^6] r_g$), with equal weight and r_0 log-uniform. M and a are frozen at their values at the injection time. The orbit migrates in a steady α -disk ($\alpha_{\text{ss}} = 0.01$, $h = 0.01$) fed at $\lambda_{\text{on}} \dot{M}_{\text{Edd}}$: Type

I ($v_r = -2C q h^{-2} \Sigma r^3 \Omega / M$, $C = 1.364 + 0.541/2$) or Type II beyond the gap-opening criterion, plus GW reaction. For these parameters the Type I time $r/|v_r| = 3\pi\alpha_{ss}h^4(M/\dot{M})/(2Cq)$ is independent of r , about 4×10^3 yr for $M = 10^6 M_\odot$, so companions reach $10^3 r_g$ well within an episode. At the end of the episode the disk is removed and the orbit evolves by GWs only, stalling if it does not reach the ISCO within 1 Gyr.

5 Resonances

Levels have

$$\frac{\omega_{nlm}}{\mu} - 1 = -\frac{\alpha^2}{2n^2} - \frac{\alpha^4}{8n^4} + \left(\frac{2}{n} - \frac{6}{2l+1}\right) \frac{\alpha^4}{n^3} + \frac{16 a m \alpha^5}{2l(2l+1)(2l+2)n^3}, \quad (2)$$

and a transition $i \rightarrow f$ is resonant when the orbital frequency equals

$$\Omega_{\text{res}} = \frac{\omega_f - \omega_i}{m_f - m_i}, \quad r_{\text{res}} = (GM/\Omega_{\text{res}}^2)^{1/3}, \quad (3)$$

with $\Omega_{\text{res}} > 0$ for orbits co-rotating with the cloud (all companions here). A transition is kept if a tidal multipole $l_* \in L$ couples it: $|l_i - l_f| \leq l_*$, $l_i + l_f + l_*$ even, $|\Delta m| \leq l_*$, and $l_* + \Delta m$ even for an equatorial orbit. Default $L = \{2, 3\}$ (`--1_stars 2` for quadrupole only). Kinds: *hyperfine* (same n, l), *fine* (same n , different l), *Bohr* (different n , final $n \leq l_i + \max L + 2$, which includes the lowest- n , i.e. outermost, line of every Δm). Only levels with $M_c/M \geq 10^{-6}$ at the crossing count. The first crossed line is recorded (lines degenerate with it are grouped); otherwise the outcome is *no cloud*, *no line ahead* or *stalled*. The bookkeeping is kinematic: no couplings or Landau-Zener probabilities, and no back-reaction on the cloud or the orbit.

For the 211 cloud with the spin near threshold, $\tilde{a} \simeq a_c(\alpha) = 4\alpha/(1 + 4\alpha^2)$, the scalings are

$$r_{\text{hf}}^{211 \rightarrow 21\bar{1}} = \left(\frac{12}{\tilde{a}\alpha^6}\right)^{2/3} r_g \propto \alpha^{-14/3}, \quad r_{\text{fine}} \propto \alpha^{-10/3}, \quad (4)$$

$$r_{\text{Bohr}} \simeq \frac{7.7}{\alpha^2} r_g \quad (211 \rightarrow 433, l_* = 2), \quad \frac{9.3}{\alpha^2} r_g \quad (211 \rightarrow 544, l_* = 3), \quad (5)$$

so the Bohr lines sit at about twice the cloud radius ($4/\alpha^2 r_g$). The hyperfine line lies at $10^6 r_g$ for $\alpha \simeq 0.06$, $10^5 r_g$ for $\alpha \simeq 0.10$ and $10^3 r_g$ for $\alpha \simeq 0.28$. Below $\alpha \approx 0.06$ every companion therefore forms inside it, and the first line it meets is Bohr; above $\alpha \approx 0.1$ in-situ companions always cross the hyperfine line first.

6 Demo: $\mu = 3 \times 10^{-17} \text{ eV}$

Twelve histories ($\alpha = 0.225$ at $10^6 M_\odot$; local α at injection 0.004–0.20), $N_{\text{max}} = 3$, 20 companions each (`src/GW_forward/analysis/test_run_forward/`):

first outcome	$L = \{2, 3\}$	$L = \{2\}$
no cloud	51.2%	51.2%
Bohr	35.0%	34.2%
hyperfine	10.0%	10.0%
fine	2.5%	2.5%
no line ahead	1.2%	2.1%

The Bohr line is mostly $|211\rangle \rightarrow |544\rangle$ for $L = \{2, 3\}$ and $|211\rangle \rightarrow |433\rangle$ for $L = \{2\}$; the hyperfine lines are $|211\rangle \rightarrow |21\bar{1}\rangle$ and $|322\rangle \rightarrow |320\rangle$ in equal numbers. *No cloud* comes from MBHs at $\alpha \leq 0.04$ (median 0.024), where the cloud has not grown. Bohr-first companions orbit MBHs at median $\alpha = 0.05$ (most within 0.04–0.07), as expected from the hyperfine radius above; hyperfine- and fine-first ones sit at $\alpha = 0.07$ –0.20 (median 0.12). Which Bohr line comes first depends on L , the Bohr share does not.

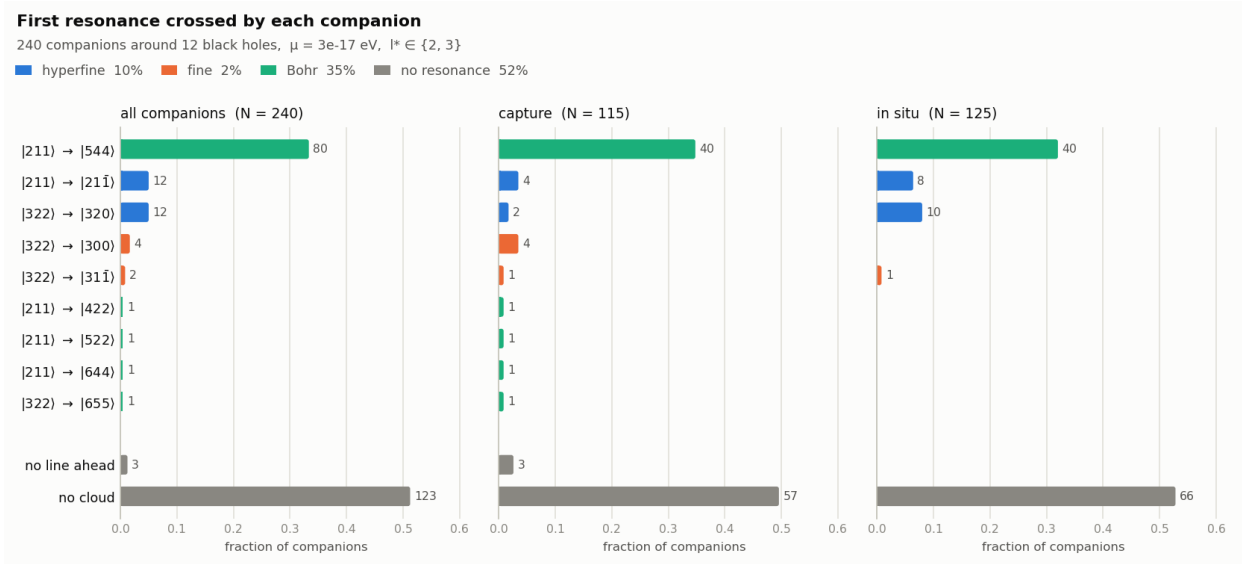


Figure 1: First transition crossed per companion in the demo run, for all companions and per formation channel.

7 Running a scan

```
cd src/GW_forward/analysis
./run_agn_scan.sh -j <cores> -N 100 -n 20 -m 4 [-q] 0.05 0.1 0.2 0.3 0.5
```

Each argument is α at $M_{\text{ref}} = 10^6 M_{\odot}$ ($-r$), one run directory per value under `scan_runs/`, with the same histories for every α (same seed). All histories share one pool of single-threaded Julia jobs; re-running the same command resumes. The output is a per-transition histogram for every α (`alpha*/first_resonance_hist.png`), plus `scan_summary.png` and `scan_kinds.csv`, `scan_transitions.csv`. The summary plots the outcome fractions against α and a `transition×α` map; `-q` repeats the analysis with quadrupole tides only. Cost in the demo: 15–40 min per history on one core at $N_{\text{max}} = 3$, several times more at $N_{\text{max}} = 4$ –5.

What to look for. The hand-over from Bohr-first to fine/hyperfine-first as α grows, and its dependence on the channel (the r_0 ranges); the switch from 211 to 322 lines; the *no cloud* fraction at small α ; and the sensitivity to L , λ_{on} and the duty cycle.

Open issues. (i) A coupling or Landau–Zener criterion to keep only lines that move population; (ii) companion back-reaction (floating, ionization); (iii) retrograde or misaligned companions (chaotic accretion is not supported by the Julia side); (iv) migration resuming in later episodes.